

	and hence the mean power dissipated across R remains unchanged.	[1]
8(a)(i)	Newton's law of gravitation states that the force of attraction between two point masses is directly proportional to the product of the masses and inversely proportional to the square of the distance between them.	[2]
8(a)(ii)	When the astronaut is in orbit, <u>gravitational force is just sufficient to provide the centripetal force</u> to keep the astronaut in circular orbit about the centre of the Earth. Hence the resultant force acting on astronaut is the gravitational force. Since <u>there is no contact force, the astronaut thus experiences weightlessness.</u>	[1] [1]

8(a)(iii)	Gravitational force provides for centripetal force. $\frac{GM_E m}{r^2} = \frac{mv^2}{r}$ $mv^2 = \frac{GM_E m}{r}$ $\frac{1}{2}mv^2 = \frac{GM_E m}{2r}$ $E_T = KE + U$ $= \frac{GM_E m}{2r} - \frac{GM_E m}{r}$ $= -\frac{GM_E m}{2r}$	[1] for statement of gforce equals centripetal force [1] for correct KE [1] for correct E_T
8(b)(i)	The term geostationary implies a satellite is fixed about a position above the Earth.	[1]
8(b)(ii)	The satellite orbit from west to east direction about Earth's axis.	[1]
8(b)(iii)	The gravitational force acting on the satellite provides the centripetal force for its circular orbit. Since the <u>gravitational force acts towards the centre of the Earth, the plane of the satellite's orbit must include the centre of the Earth.</u> <u>For it to be geostationary, the axis of orbit of satellite must coincide with axis of rotation of Earth.</u> <u>Since plane of rotation is always perpendicular to axis of rotation, satellite must be above the equator.</u>	[1] [1] [1]
8(b)(iv)	$v = r\omega$ $= 4.23 \times 10^7 \times \frac{2\pi}{24 \times 3600}$ $= 3080 \text{ m s}^{-1}$	[1] correct substitution [1] correct answer
8(b)(v)	$\frac{GM_E m}{r^2} = \frac{mv^2}{r}$ $GM_E = v^2 r$ $= (3076)^2 (4.23 \times 10^7)$ $GM_E = 4.0027 \times 10^{14}$ $\Delta U = -\frac{GM_E m}{r_f} - \left(-\frac{GM_E m}{r_i} \right)$ $\Delta U = -\frac{4.0027 \times 10^{14} \times 2400}{4.23 \times 10^7} - \left(-\frac{4.0027 \times 10^{14} \times 2400}{6.38 \times 10^6} \right)$ $\Delta U = 1.28 \times 10^{11} \text{ J}$	[1] for correct GM_E [1] correct substitution [1] correct answer

8(b)(vi)	Let the minimum speed of projection be v_0. K.E. of satellite at orbit $= \frac{1}{2} \times 2400 \times \left(4.23 \times 10^7 \times \frac{2\pi}{24 \times 3600} \right)^2$ $= 1.136 \times 10^{10} \text{ J}$ When the satellite is projected from the Equator, Loss in K.E. = gain in G.P.E. Initial K.E. at the Earth's surface – K.E. at orbit = gain in G.P.E $\frac{1}{2}mv_o^2 - \frac{1}{2}mv_{orbit}^2 = 1.28 \times 10^{11}$ $\frac{1}{2}mv_o^2 - 1.136 \times 10^{10} = 1.28 \times 10^{11}$ $v_o = 1.08 \times 10^4 \text{ m s}^{-1}$	[1] KE at orbit [1] correct substitution [1] correct answer
8(c)	The work function energy of a surface is the minimum amount of	[1]